📘 IB DIPLOMA PROGRAMME

IB Environmental Systems & Societies

Complete Cheat Sheet

Full Syllabus Coverage · Exam-Ready Revision · Score a 7
6 Major Topics · All Key Cycles · Case Studies Included

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7
Target Score

- 📄 20 A4 Pages
- 🧠 High-Density Content
- ⚡ Fast Revision
- 🎯 Exam-Focused

Foundations of ESS | Ecosystems & Ecology | Biodiversity & Conservation | Water & Food Systems | Atmospheric & Climate | Human Systems & Resources

How to Use This Cheat Sheet

STUDY GUIDE

Using This Document

1. Read each section before your topic exam
2. Use **Exam Tips** to prioritise high-yield content
3. Study the **diagrams** — IB loves visual questions
4. Practise **evaluation** using pros/cons boxes
5. Check **Common Mistakes** (p.17) before every exam
6. Use the **Night Before Checklist** (p.20) on exam eve

EXAM TIP: IB ESS is 50% Paper 1 (case study) and 50% Paper 2 (essays + data). This sheet covers BOTH.

Icon Legend

- **Exam Tip** — high-yield exam strategy
- **Exam Insight** — what IB commonly tests
- **Common Mistake** — avoid this error
- **Case Study** — real-world example
- **Calculation** — formula to know

FULL SYLLABUS MAP

Topic	Pages	Key Focus
1 · Foundations of ESS	3–4	Systems, sustainability, models
2 · Ecosystems & Ecology	5–6	Energy, food webs, productivity
3 · Biodiversity & Conservation	7–8	Threats, strategies, evaluation
4 · Water & Food Production	9–11	Scarcity, agriculture, sustainability
5 · Atmospheric & Climate	12–14	GHGs, impacts, mitigation
6 · Human Systems & Resources	15–16	Consumption, waste, strategy
Exam Mastery	17–20	Tips, commands, diagrams, checklist

Assessment Overview

Component	Weight	What's Tested
Paper 1	25%	Unseen case study (data response)
Paper 2	50%	Short answer + 2 extended essays
IA	25%	Investigation (individual project)

EXAM INSIGHT: Paper 2 essays ALWAYS require evaluation (pros/cons, perspectives, sustainability). Practice this structure: State → Explain → Example → Evaluate.

TOPIC CONNECTIONS

Foundations Systems + Models → **Ecosystems** Energy + Food → **Biodiversity** Conservation → **Water & Food** Agriculture → **Atmosphere** Climate → **Human Systems** Resources

Foundations of ESS — Core Concepts

SYSTEMS

What is a System?

A set of interacting components that form a unified whole. Systems have **inputs**, **outputs**, **storages**, and **flows**.

Type	Matter	Energy	Example
Open	Exchanges	Exchanges	Tropical rainforest
Closed	No exchange	Exchanges	Earth (approx.)
Isolated	No exchange	No exchange	Theoretical only

Key System Terms

- **Storage/Stock:** amount of material in a system at a given time
- **Flow/Flux:** rate of transfer between stores
- **Input:** matter/energy entering the system
- **Output:** matter/energy leaving the system
- **Throughput:** matter/energy passing through

Feedback Loops

- **Negative feedback:** dampens change, promotes stability (thermostat model) → homeostasis
- **Positive feedback:** amplifies change, destabilises system → e.g. ice-albedo feedback

💡 **EXAM TIP:** IB commonly asks you to identify feedback types from diagrams. Negative = stabilising. Positive = amplifying.

SUSTAINABILITY

Sustainability Definitions

- **Sustainable development:** "Meeting the needs of the present without compromising the ability of future generations to meet their own needs" (Brundtland, 1987)
- **Natural capital:** stocks of natural resources (forests, water, minerals)
- **Natural income:** sustainable yield from natural capital

Sustainability Spectrum

View	Description
Ecocentric	Nature has intrinsic value; human needs subordinate to ecosystem health
Technocentric	Technology can solve environmental problems; human innovation is key
Anthropocentric	Humans are central; environment valued for human benefit

ENERGY VS NUTRIENTS

Energy Flow

- ONE-WAY flow
- Lost as heat at each trophic level
- Non-renewable within ecosystem
- Source: Sun (photosynthesis)

Nutrient Cycling

- CYCLICAL movement
- Matter recycled continuously
- Involves biotic + abiotic components
- Key cycles: C, N, P, H_2O

⚠️ **COMMON MISTAKE:** Confusing energy flow (one-way) with nutrient cycling (cyclical) is one of the most common errors in Paper 2!

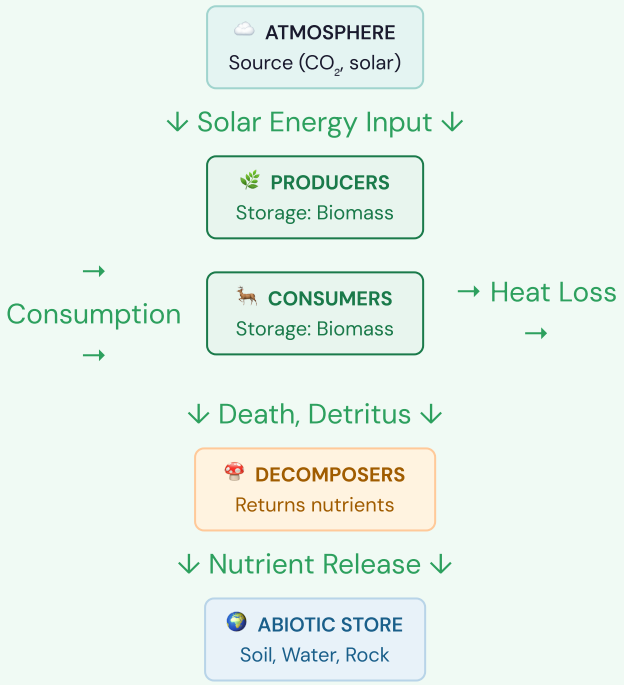
Foundations of ESS — Systems Models

SYSTEM DIAGRAMS

Components of a System Diagram

- **Box (rectangle):** storage/stock
- **Arrow (solid):** flow/flux between storages
- **Valve symbol:** rate of flow (tap)
- **Cloud:** source or sink (outside system boundary)

SIMPLE ECOSYSTEM SYSTEM DIAGRAM



EQUILIBRIUM & CHANGE

Concept	Definition
Steady state	Inputs = Outputs; system remains constant over time
Dynamic equilibrium	Constant fluctuation around a balance point
Tipping point	Threshold beyond which system shifts to new state irreversibly
Resilience	Ability of system to recover from disturbance
Lag time	Delay between cause and observable effect

EVS FRAMEWORKS

Environmental Value Systems (EVS)

EVS	Core Belief	Example Position
Ecocentric	Deep ecology; intrinsic value in nature	Protect all species regardless of utility
Anthropocentric	Humans are stewards of nature	Use resources sustainably for human benefit
Technocentric	Science & technology can fix problems	Nuclear power, GMOs, geoengineering as solutions

EXAM INSIGHT: EVS questions ask you to argue from DIFFERENT perspectives. Show you can think like an ecocentrist AND a technocentrist.

Ecological Footprint

- Measures human demand on natural capital in global hectares (gha)
- Compares resources used vs Earth's biocapacity
- **Overshoot:** demand exceeds regeneration rate
- **Earth Overshoot Day:** point when annual demand exceeds yearly supply

COMMON MISTAKE: Don't confuse *ecological footprint* (area of land) with *carbon footprint* (tonnes CO_2). IB marks on precision!

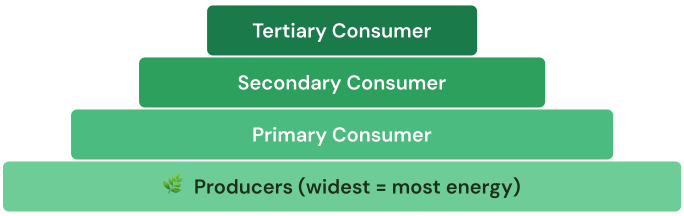
Ecosystems & Ecology — Food Webs & Energy

FOOD CHAINS & WEBS

Term	Definition
Producer	Autotroph; makes own food via photosynthesis/chemosynthesis (e.g. grass, algae)
Primary consumer	Herbivore; eats producers (e.g. rabbit, caterpillar)
Secondary consumer	Eats primary consumers (e.g. fox, frog)
Tertiary consumer	Eats secondary consumers (e.g. eagle, shark)
Decomposer	Breaks down dead organic matter; returns nutrients to soil (bacteria, fungi)
Detritivore	Feeds on detritus (e.g. earthworm, woodlouse)

💡 **EXAM TIP:** Food *web* = multiple interacting food chains. Always use food web, not just food chain, in extended answers.

ENERGY PYRAMIDS



10% Rule (Lindeman's Efficiency)

Only ~**10%** of energy at one trophic level is transferred to the next. ~90% is lost as heat (respiration), excretion, or non-consumed biomass.

Energy efficiency = (Energy at T_{n+1} / Energy at T_n) × 100

TROPHIC LEVELS

Trophic Level Summary

Level	Name	Example
T1	Producers	Grasses, phytoplankton, trees
T2	Primary consumers	Rabbits, zooplankton, insects
T3	Secondary consumers	Foxes, small fish, frogs
T4	Tertiary consumers	Eagles, large predators
T5	Apex predators	Orca, lion, humans (sometimes)

BIOMASS PYRAMIDS

Types of Ecological Pyramids

Type	Y-axis	Can invert?
Energy	Energy (kJ/m²/yr)	Never inverts
Biomass	Dry mass (g/m²)	Can invert (aquatic: phytoplankton < zooplankton)
Numbers	Count of organisms	Can invert (trees < insects)

🎯 **EXAM INSIGHT:** IB often asks: "Why is energy pyramid always upright?" → Because energy is lost at each level; it can never increase up the pyramid.

Ecological Efficiency Implications

- Eating lower trophic levels = **more efficient use** of food energy
- Vegetarian diets require ~10x less land than meat-based diets
- Short food chains = more energy available to humans

🌍 **CASE STUDY:** Amazon rainforest: complex food webs with up to 5 trophic levels. Deforestation collapses entire food webs by removing producer base.

Ecosystems — Productivity & Limiting Factors

PRIMARY PRODUCTIVITY

Key Definitions

- **GPP (Gross Primary Productivity):** total energy fixed by producers via photosynthesis per unit area per time (kJ/m²/yr)
- **NPP (Net Primary Productivity):** energy available to next trophic level after plant respiration
- **Respiration (R):** energy used by producers for their own metabolic processes

NPP = GPP − Respiration

Net biomass available for consumption

Typical NPP Values

Ecosystem	NPP (gC/m²/yr)
Tropical rainforest	800–3,500 (highest)
Temperate forest	400–2,000
Grassland	200–1,000
Open ocean	2–400 (low)
Desert	0–40 (lowest)

💡 **EXAM TIP:** NPP questions often ask you to calculate. Know the formula and be able to convert units (kJ → gC: 1gC = ~45 kJ).

LIMITING FACTORS

Liebig's Law of the Minimum

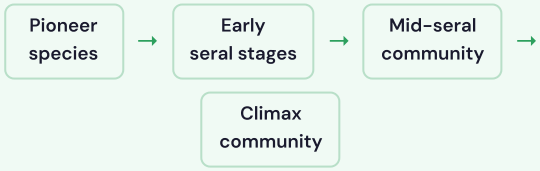
Growth is limited by the **scarcest essential resource**, not by the total of all resources.

Factor	Effect on Productivity
Light	Powers photosynthesis; too little → low GPP; too much → photorespiration
CO_2 conc.	Substrate for photosynthesis; higher CO_2 → faster fixation (up to a point)
Water	Reactant in photosynthesis; deficit causes stomatal closure
Nutrients (N, P)	Essential for growth; excess causes eutrophication
Temperature	Affects enzyme activity; optimal range for each species
Grazing/herbivory	Reduces standing crop of producers

SUCCESSION

Ecological Succession

Type	Starts from	Example
Primary	Bare/lifeless substrate	Lava flows, sand dunes, glacial retreat
Secondary	Disturbed land (soil intact)	Abandoned farmland, after fire



🎯 **EXAM INSIGHT:** Succession questions: name pioneer species (e.g. lichens, mosses), describe soil formation, and explain how biodiversity changes at each stage (increases towards climax).

Biodiversity & Conservation

TYPES OF BIODIVERSITY

Three Levels

- **Genetic diversity:** variety of alleles/genes within a species → enables adaptation
- **Species diversity:** number and relative abundance of species in an area (richness + evenness)
- **Ecosystem diversity:** variety of different habitats/ecosystems in a region

Simpson's Diversity Index (D)

$$D = 1 - \Sigma[n(n-1) / N(N-1)]$$

- n = number of individuals of each species
- N = total number of individuals of all species
- D closer to 1 = higher diversity
- D closer to 0 = lower diversity

EXAM TIP: Always show your working in Simpson's calculation. IB awards process marks even if your final answer is wrong!

THREATS TO BIODIVERSITY

HIPPO Mnemonic

- **H** **Habitat loss/fragmentation** — #1 cause globally
- **I** **Invasive species** — compete, predate, exclude natives
- **P** **Pollution** — air, water, soil, noise, light
- **P** **Population growth** — human demand on resources
- **O** **Overexploitation** — overfishing, hunting, logging
- **+** **Climate change** — shifting ranges, phenology mismatch

CONSERVATION STRATEGIES

Strategy	Method	Example
In-situ	Conservation in natural habitat	National parks, marine reserves, wildlife corridors
Ex-situ	Conservation outside habitat	Zoos, seed banks, captive breeding
Legislation	Legal protection of species/habitat	CITES, Ramsar Convention, national laws
Community	Local stewardship	Community reserves, payments for ecosystem services

✅ In-situ Advantages

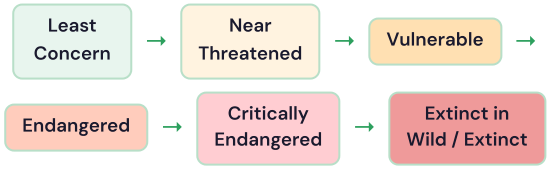
- Natural behaviors maintained
- Entire ecosystems protected
- Lower cost per species
- Preserves species interactions

❌ In-situ Disadvantages

- Ongoing threats not removed
- Small area = inbreeding risk
- Political/land-use conflicts
- May miss critically endangered

CASE STUDY: Virunga National Park (DRC): in-situ conservation of mountain gorillas (~1,000 remain. Success due to ranger protection + ecotourism revenue. Threat: armed conflict + charcoal production.

IUCN Red List Categories



EXAM INSIGHT: Evaluation of conservation: Always compare in-situ vs ex-situ. Mention costs, effectiveness, feasibility, and who benefits (stakeholders).

Biodiversity — Ecosystem Services & Evaluation

ECOSYSTEM SERVICES

Type	Example
Provisioning	Food, timber, freshwater, medicines
Regulating	Climate regulation, flood control, pollination, water purification
Cultural	Recreation, tourism, spiritual value, education
Supporting	Nutrient cycling, soil formation, primary production (underpins all others)

EXAM INSIGHT: TEEB (The Economics of Ecosystems and Biodiversity) estimated nature provides $125–145 trillion/yr in services. Use this to argue economic case for conservation.

BIODIVERSITY VALUE ARGUMENTS

Reasons to Conserve Biodiversity

- **Intrinsic value:** all species have a right to exist (ecocentric)
- **Instrumental value:** species provide resources for humans
- **Option value:** potential future uses (medicines, genes)
- **Existence value:** satisfaction from knowing species exist

GENETICS & CONSERVATION

Minimum Viable Population (MVP)

- Smallest population size that can persist long-term
- Genetic bottleneck: low diversity → inbreeding depression
- Allee effect: very small populations have reduced reproductive success
- Metapopulation: network of subpopulations linked by migration

CASE STUDIES

CASE STUDY: **Millennium Seed Bank (Kew, UK):** Ex-situ conservation — stores 2.4 billion seeds from 40,000+ species. Aims to bank 25% of world's plant species by 2020. Effective for plants, not animals.

CASE STUDY: **Coral Triangle (SE Asia):** World's most biodiverse marine ecosystem. Threats: coral bleaching, overfishing, coastal development. Conservation: MPAs (marine protected areas), community fisheries management.

CASE STUDY: **Rewilding — Yellowstone wolves:** Reintroduction of wolves (1995) triggered trophic cascade: controlled elk → riverbanks recovered → beavers returned → altered river courses. Demonstrates top-down ecosystem effects.

INTERNATIONAL AGREEMENTS

Agreement	Focus	Year
CBD	Convention on Biological Diversity — overall biodiversity framework	1992
CITES	Regulates trade in endangered species (3 appendices)	1973
Ramsar	Protection of internationally important wetlands	1971
CMS	Conservation of migratory species	1979
Kunming-Montreal	"30x30" — protect 30% of land/sea by 2030	2022

COMMON MISTAKE: Don't say species are "important" without specifying WHY (intrinsic, economic, ecological). IB requires justified reasoning.

Water Systems & Water Scarcity

THE HYDROLOGICAL CYCLE

WATER CYCLE — KEY FLOWS

1. **Evaporation** — Solar energy evaporates surface water
2. **Transpiration** — Plants release vapour (evapo-transpiration)
3. **Condensation** — Vapour cools, forms clouds
4. **Precipitation** — Rain, snow, sleet falls to surface
5. **Surface Runoff** — Water flows over land to rivers
6. **Infiltration** — Water enters soil → groundwater
7. **Groundwater** — Stored in aquifers; slow release
8. **Collection** — Returns to ocean, lakes, rivers

Water Distribution

- **97.5%** of Earth's water = saltwater (oceans)
- **2.5%** = freshwater; of which:
- **~69%** locked in glaciers/ice
- **~30%** groundwater
- **<1%** accessible surface freshwater

WATER SCARCITY

Type	Cause	Region
Physical scarcity	Insufficient rainfall/water supply	Middle East, Sahel, Australian interior
Economic scarcity	Lack of infrastructure despite water presence	Sub-Saharan Africa

Causes of Water Scarcity

- Population growth → increased demand
- Agriculture = 70% of global freshwater use (irrigation)
- Industrial use (cooling, processing)
- Pollution of water sources
- Climate change → altered precipitation patterns
- Overextraction of aquifers (e.g. Ogallala Aquifer, USA)

WATER MANAGEMENT

Strategy	Description	Example
Dams/reservoirs	Store water, generate hydroelectricity	Three Gorges Dam, China
Desalination	Remove salt from seawater	Saudi Arabia, Israel, UAE
Water transfer	Move water from surplus to deficit areas	South-North Water Transfer, China
Drip irrigation	Targeted delivery; 90% efficiency	Israel's agricultural sector
Water recycling	Treat and reuse wastewater	Singapore's NEWater
Groundwater management	Regulate aquifer extraction rates	EU Water Framework Directive

CASE STUDY: Aral Sea: Irrigation diversions from Amu Darya/Syr Darya rivers for Soviet cotton production caused 90% volume loss by 2007. Mass biodiversity loss, salinization, local climate change, public health impacts. Partial restoration in North Aral Sea via Dike Kokaral (Kazakhstan).

Food Production Systems

AGRICULTURE TYPES

System	Features	Impact
Subsistence	Small-scale, local consumption, low input	Low environmental impact; low yield
Intensive	High inputs (fertilisers, pesticides), mechanised, monocultures	High yield; soil degradation, pollution, GHG emissions
Extensive	Large area, low input per unit area, often pastoral	Habitat conversion; lower yield/ha
Organic	No synthetic chemicals, natural pest control	Higher price; better soil health, biodiversity
Urban farming	Rooftop gardens, vertical farms, hydroponics	Reduced transport, water-efficient; limited scale

GREEN REVOLUTION

✅ Advantages

- Doubled grain yields 1960–90
- Fed rapidly growing population
- Reduced food prices
- HYV crops widely available

❌ Disadvantages

- Reduced genetic diversity
- Groundwater overuse
- Pesticide/fertiliser pollution
- Ignored poor farmers (cost)
- Soil degradation

AQUACULTURE VS WILD FISHING

Feature	Wild Fishing	Aquaculture
Control	Low — natural stocks	High — controlled production
Environmental impact	Bycatch, habitat damage	Pollution, disease, escapees
Food security	Vulnerable to overexploitation	More predictable supply
Efficiency	Low — indirect energy use	Higher for some species

🌍 **CASE STUDY:** Atlantic cod collapse (Grand Banks, 1990s): Overfishing depleted cod to <1% of 1960s levels. Moratorium declared 1992. Stocks have not fully recovered. Demonstrates the tragedy of the commons in open-access fisheries.

SOIL DEGRADATION

Causes & Consequences

- **Salinisation:** irrigation → salt accumulation → loss of soil fertility
- **Erosion:** wind/water removes topsoil (bare fields, deforestation)
- **Compaction:** heavy machinery → reduced infiltration
- **Nutrient depletion:** intensive crops exhaust soil nutrients
- **Acidification:** fertiliser overuse, acid rain

💡 **EXAM TIP:** For any food system question: discuss energy efficiency, environmental impact, social equity, AND economic viability. IB rewards multi-perspective evaluation.

Sustainability of Food & Water Systems

SUSTAINABLE FOOD PRODUCTION

Strategy	Description	Benefit
Agroforestry	Trees + crops together	Soil health, carbon storage, biodiversity
Crop rotation	Alternate crops seasonally	Prevents nutrient depletion, breaks pest cycles
Integrated Pest Mgt	Biological + chemical controls	Reduces pesticide use, maintains predators
Permaculture	Designed to mimic natural ecosystems	Self-sustaining, minimal waste
Precision agriculture	GPS, sensors to optimise inputs	Reduced fertiliser, water, emissions
Reduced meat diet	Less land, water, energy needed	Lower GHG, frees land for other uses

🎯 **EXAM INSIGHT:** Sustainability questions need BOTH environment AND human dimensions. Always mention food security, equity, and feasibility for different groups.

🌍 **CASE STUDY:** Cuba's "Special Period" (1990s): loss of Soviet oil forced transition to organic/low-input farming. Urban agriculture expanded. Showed alternatives to industrial farming can work with political will.

SUSTAINABILITY FRAMEWORKS

Three Pillars of Sustainability

🌿 **Environmental**
Ecosystem health, resource limits

💰 **Economic**
Viability, efficiency, growth

👥 **Social**
Equity, access, wellbeing

Planetary Boundaries (Rockström et al.)

- 9 Earth system processes with "safe operating space"
- **Already crossed:** climate change, biodiversity loss, land-system change, biogeochemical flows (N, P), freshwater change, novel entities
- Provides scientific basis for sustainability policy

WATER SECURITY

Indicator	Threshold
Water stress	<1,700 m³/person/year
Water scarcity	<1,000 m³/person/year
Absolute scarcity	<500 m³/person/year

⚠️ **COMMON MISTAKE:** Don't forget virtual water: food production requires large hidden water amounts. 1 kg beef ≈ 15,400 litres of water. Food trade = water trade!

Atmospheric Systems & Greenhouse Effect

ATMOSPHERIC STRUCTURE

Layer	Altitude	Feature
Troposphere	0–12 km	Weather, CO_2, most GHGs, temperature ↓ with altitude
Stratosphere	12–50 km	Ozone layer (15–35 km), temperature ↑ with altitude
Mesosphere	50–85 km	Meteors burn up; temperature ↓ with altitude
Thermosphere	85–600 km	Auroras; very thin; temperature ↑ with altitude

THE GREENHOUSE EFFECT

NATURAL GREENHOUSE EFFECT

1. Short-wave solar radiation reaches Earth
2. Earth's surface absorbs energy & warms
3. Long-wave infrared radiation re-emitted upward
4. GHGs absorb IR radiation, re-emit in all directions

→ Earth's surface warms (natural effect = +33°C, essential for life)

Greenhouse Gases (GHGs)

Gas	GWP (100yr)	Key Source
CO_2	1 (baseline)	Fossil fuels, deforestation, cement
CH_4 (methane)	28–36	Livestock, landfill, rice paddies, gas leaks
N_2O	265–298	Agriculture (fertilisers), combustion
HFCs/CFCs	Thousands	Refrigerants, aerosols (also ozone depleting)
H_2O vapour	N/A	Natural; amplifies warming (positive feedback)

ENHANCED GREENHOUSE EFFECT

Human Enhancement of GHG

- Burning fossil fuels → more CO_2 released than natural sinks can absorb
- Deforestation → less CO_2 absorbed + stored carbon released
- Agriculture → CH_4 from livestock & rice, N_2O from fertilisers
- Industrial processes → fluorinated gases
- Landfill → CH_4 from decomposing waste

OZONE DEPLETION

Ozone Layer & CFCs

- Ozone (O_3) layer in stratosphere absorbs UV-B radiation
- CFCs (chlorofluorocarbons) break down ozone catalytically
- 1 Cl atom destroys ~100,000 O_3 molecules
- Antarctic ozone hole first detected 1985
- **Montreal Protocol (1987):** banned CFCs → ozone recovering

CASE STUDY: Montreal Protocol success: CFC phase-out led to measurable ozone recovery. Antarctic ozone hole expected to close by ~2060. Considered one of the most successful international environmental agreements.

EXAM INSIGHT: IB distinguishes between ozone depletion (different problem!) and climate change. CFCs cause both. Show you understand the difference.

COMMON MISTAKE: Ozone depletion ≠ climate change. They are related but distinct problems with different causes, mechanisms, and effects. IB marks on this distinction!

Climate Change — Causes & Impacts

EVIDENCE FOR CLIMATE CHANGE

Evidence	Finding
Temperature records	+1.1°C average surface warming since 1850–1900 (IPCC AR6, 2021)
Ice cores	CO_2 and temperature co-vary over 800,000 years; current CO_2 highest in record
Sea level rise	~20 cm since 1900; accelerating due to thermal expansion + ice melt
Arctic sea ice	~40% reduction in summer extent since 1979
Glacier retreat	Virtually all monitored glaciers retreating globally
Species range shifts	Species moving poleward/upslope; phenological changes

KEY FEEDBACK MECHANISMS

Positive Feedbacks (Amplifiers)

- **Ice-albedo:** ice melts → darker ocean absorbs more heat → more warming
- **Water vapour:** warmer air holds more water vapour (GHG) → more warming
- **Permafrost thaw:** releases CH_4 + CO_2 → accelerates warming
- **Forest dieback:** Amazon at risk of savannification; releases stored C

Negative Feedbacks (Stabilisers)

- **Plant growth:** more CO_2 → more photosynthesis → absorbs some CO_2
- **Planktonic blooms:** warmer oceans → some bloom → CO_2 uptake
- **Increased cloud cover:** more evaporation → more clouds → reflects solar radiation

IMPACTS OF CLIMATE CHANGE

Category	Key Impacts
Physical	Sea level rise, more frequent extreme weather, changing precipitation, glacier retreat
Ecological	Coral bleaching, species range shifts, phenology mismatch, habitat loss, mass extinctions
Agricultural	Crop yield changes, water stress, new pests, altered growing seasons
Human health	Heat stress, vector-borne disease spread (malaria, dengue), malnutrition
Socioeconomic	Climate refugees, coastal infrastructure damage, water conflicts, economic costs

CASE STUDY: Great Barrier Reef: 50% of coral cover lost since 1995 due to bleaching events (2016, 2017, 2020, 2022 — mass bleaching events). Root cause: elevated sea surface temperatures from climate change. Compounded by water quality and crown-of-thorns starfish.

IPCC Scenarios (RCPs/SSPs)

- **SSP1-1.9:** Net zero by 2050 → +1.5°C (best case)
- **SSP2-4.5:** Moderate mitigation → +2.7°C
- **SSP5-8.5:** Business as usual → +4.4°C by 2100

EXAM INSIGHT: IPCC reports are the gold standard IB reference. Mention IPCC AR6 (2021/22) in essays on climate change — shows awareness of current scientific consensus.

Climate Change — Mitigation & Adaptation

MITIGATION STRATEGIES

Strategy	How it works	Evaluation
Renewable energy	Solar, wind, hydro replace fossil fuels	+Clean; -Intermittent, land use, upfront cost
Energy efficiency	Better insulation, LED, EVs	+Cost-saving; -Rebound effect risk
Carbon capture (CCS)	Capture CO_2 at source, store underground	+Large-scale potential; -Expensive, unproven at scale
Reforestation	Planting trees to sequester CO_2	+Multiple co-benefits; -Time lag, land competition
Carbon trading	Cap-and-trade market for emissions	+Market mechanism; -Price instability, fraud risk
Carbon tax	Price on carbon emissions	+Simple, effective; -Regressive, politically difficult

🎯 **EXAM INSIGHT:** IB essays on mitigation require evaluation of at least 2 strategies with pros AND cons. Compare economic vs regulatory approaches.

ADAPTATION STRATEGIES

Hard vs Soft Adaptation

Hard Engineering	Soft/Managed
Sea walls, flood barriers	Managed retreat from coastlines
Desalination plants	Drought-resistant crop varieties
Air conditioning expansion	Urban green spaces (cooling)

INTERNATIONAL CLIMATE AGREEMENTS

Agreement	Year	Key Commitment
Kyoto Protocol	1997	Binding emission targets for developed nations (1st period 2008–12)
Copenhagen Accord	2009	Non-binding; limit warming to 2°C
Paris Agreement	2015	NDCs; limit warming to 1.5–2°C; net zero by 2050
COP28 (Dubai)	2023	Transition away from fossil fuels; loss and damage fund

✅ Paris Agreement Strengths

- Universal participation (195 parties)
- Flexible NDC approach
- 5-yr ratchet mechanism
- Climate finance for developing nations

❌ Paris Agreement Weaknesses

- NDCs are voluntary
- No enforcement mechanism
- Current pledges → ~2.7°C
- USA withdrew under Trump

⚠️ **COMMON MISTAKE:** Don't say "Paris Agreement solved climate change." Current pledges are insufficient for 1.5°C. Show awareness of the gap between pledges and action.

Human Systems & Resource Use

URBANISATION

Urban Environmental Impacts

- **Urban Heat Island (UHI):** cities warmer than surroundings due to concrete, vehicles, lack of vegetation
- **Impervious surfaces:** increased runoff, flash flooding, reduced groundwater recharge
- **Air pollution:** NO_x, particulates, ozone from transport, industry
- **Waste generation:** concentrated waste streams require management infrastructure

FOSSIL FUELS

Benefits

- High energy density
- Existing infrastructure
- Reliable base-load power
- Economic development

Costs

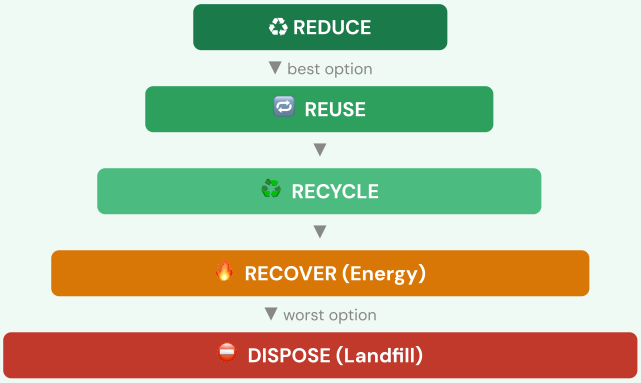
- GHG emissions
- Air/water pollution
- Non-renewable
- Price volatility
- Geopolitical conflicts

ENERGY RESOURCES

Type	Examples	Renewable?
Fossil fuels	Coal, oil, gas	No
Nuclear	Uranium fission	Debated
Renewables	Solar, wind, hydro, geothermal, tidal, biomass	Yes

WASTE MANAGEMENT

THE WASTE HIERARCHY



Pollution Types & Management

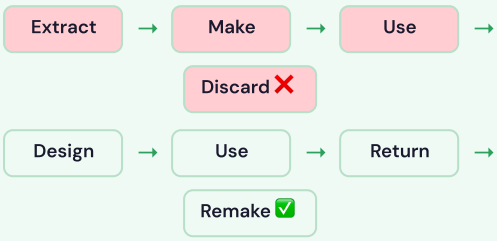
Type	Source	Management
Eutrophication	N/P runoff from agriculture	Nutrient management, wetland buffers
Acid rain	SO_2, NO_x from combustion	Scrubbers, catalytic converters, fuel switching
Bioaccumulation	Persistent pollutants (DDT, mercury)	Ban, substitution, phytoremediation
Plastic pollution	Packaging, microplastics	Bans, extended producer responsibility

CASE STUDY: Curitiba, Brazil: model sustainable city — extensive BRT system, green space integration, recycling programs, flood management. Often cited as example of urban sustainability planning.

Sustainability Strategies & Global Solutions

CIRCULAR ECONOMY

Linear vs Circular



RENEWABLE ENERGY TRANSITION

Source	Pros	Cons
Solar PV	Scalable, falling costs, distributed	Intermittent, land use, panel disposal
Wind	Low lifecycle emissions, offshore potential	Visual impact, bird mortality, intermittent
Hydroelectric	Reliable, long lifespan, storage (pumped)	Habitat disruption, displacement, methane (reservoirs)
Nuclear	Low carbon, reliable baseload, high energy density	Waste storage, accident risk, high cost, public fear
Geothermal	Constant output, low land use	Location-limited, H_2S emissions

SDGS & GLOBAL GOVERNANCE

UN Sustainable Development Goals (2030 Agenda)

- 17 goals adopted 2015, 169 targets
- Relevant ESS goals: SDG 6 (clean water), SDG 7 (energy), SDG 11 (sustainable cities), SDG 13 (climate action), SDG 14 (life below water), SDG 15 (life on land)
- Framework for evaluating sustainability policies

EVALUATION: SOLUTIONS COMPARISON

Approaches to Environmental Problems

Approach	Example	Strength	Weakness
Regulatory	Emission standards, bans	Clear targets	Enforcement costs
Economic	Carbon tax, subsidies	Market efficiency	Equity concerns
Technological	CCS, EVs, GMOs	Scalable solutions	Risk, cost, time lag
Behavioural	Education, nudge theory	Low cost, long-term	Slow, uncertain effect

CASE STUDY: Germany's Energiewende (energy transition): 46% renewable electricity by 2023. Strong policy + subsidies drove growth. Challenges: grid stability, coal phase-out timeline, high consumer electricity prices.

EXAM INSIGHT: When asked to evaluate a solution, use this structure: Define → Explain mechanism → Provide example → Evaluate (pros/cons/limitations) → Conclusion. This earns maximum marks.

Exam Mastery — Common Mistakes

TOP 15 IB ESS EXAM MISTAKES

⚠ COMMON MISTAKE: **1. Energy vs Nutrient Flow:** Saying energy "cycles" — energy flows ONE WAY and is lost as heat. Only nutrients cycle.

⚠ COMMON MISTAKE: **2. GPP vs NPP:** Using GPP and NPP interchangeably. NPP = GPP - Respiration. NPP is available to next trophic level.

⚠ COMMON MISTAKE: **3. Ozone ≠ Climate Change:** Treating ozone depletion and enhanced greenhouse effect as the same problem. Different gases, mechanisms, effects.

⚠ COMMON MISTAKE: **4. "Biodiversity" without specificity:** Saying "biodiversity decreases" without stating WHICH type (species, genetic, ecosystem) or HOW.

⚠ COMMON MISTAKE: **5. Inverted pyramid confusion:** Claiming all pyramids can invert. Only biomass and numbers can invert — energy pyramids NEVER invert.

⚠ COMMON MISTAKE: **6. Carbon footprint ≠ Ecological footprint:** Carbon footprint measures CO_2 emissions. Ecological footprint measures land area required for resource use.

⚠ COMMON MISTAKE: **7. Succession endpoint:** Calling climax community "most stable" without noting it depends on disturbance regime. Fire-climax communities exist.

⚠ COMMON MISTAKE: **8. Ignoring feedback loops:** Describing climate change impacts without mentioning feedback mechanisms (positive = amplifying, negative = stabilising).

⚠ COMMON MISTAKE: **9. Vague evaluation:** Saying a strategy is "good for the environment." Always specify what aspect, to what extent, with what trade-offs.

⚠ COMMON MISTAKE: **10. Command term neglect:** "Describing" when asked to "evaluate" or "explain." Always re-read the command term. IB marks specifically to command level.

⚠ COMMON MISTAKE: **11. No case study in extended answers:** Extended essays require at least ONE specific, named real-world example. Generic answers score max 6/9.

⚠ COMMON MISTAKE: **12. Confusing in-situ/ex-situ:** In-situ = in the wild (national park). Ex-situ = removed from habitat (zoo, seed bank). Don't reverse these.

⚠ COMMON MISTAKE: **13. Ignoring social/economic dimensions:** Environmental solutions must consider feasibility, equity, and human wellbeing — not just ecology.

⚠ COMMON MISTAKE: **14. Mixing up Simpson's formula:** Writing $D = \Sigma n(n-1)/N(N-1)$ without the "1-" prefix. The full formula is $D = 1 - \Sigma[n(n-1)/N(N-1)]$.

⚠ COMMON MISTAKE: **15. Unsupported claims:** Making statements without scientific evidence or reasoning. IB expects evidence-based argument, not opinion.

Exam Strategy & Command Terms

IB COMMAND TERMS

Give account of similarities AND differences

Term	What IB expects	Marks
Define	Give precise meaning of a term	1–2
State	Give specific name, value, or fact (no explanation)	1
Outline	Brief summary of key points; some explanation	2–3
Describe	Give detailed account of features; no justification needed	2–4
Explain	Describe + give reasons/mechanisms (how/why)	3–5
Discuss	Consider multiple perspectives; balanced analysis	6–9
Evaluate	Assess merits, limitations, significance; reach judgement	6–9
Justify	Give valid reasons or evidence to support an answer	2–4
Suggest	Propose possible answers (more than one acceptable)	1–2
Compare	3–5	
Construct	Transform data into a diagram/graph/model	3–5

ESSAY STRUCTURE (7-SCORE TEMPLATE)

Extended Essay Framework (9-mark questions)

- P1 **Define** key terms from the question
- P2 **Identify** the main argument/system
- P3 **Explain** mechanism with detail and science
- P4 **Example** — specific, named, real-world case
- P5 **Evaluate** — pros AND cons, multiple perspectives
- P6 **Consider** EVS (ecocentric vs technocentric)
- P7 **Conclude** — nuanced judgement, acknowledge complexity

TIME MANAGEMENT

Paper	Time	Strategy
Paper 1	1h	Read case study twice. Answer data questions first. Allocate 10 min/mark.
Paper 2 SA	~1h	2 min/mark. Skip & return if stuck. Always attempt every part.
Paper 2 Essay	20 min each	5 min plan → 12 min write → 3 min review. Include diagram if relevant.

EXAM TIP: Diagrams score marks! A labelled diagram in an essay question shows understanding and saves words. Always label axes, arrows, stores clearly.

EXAM INSIGHT: Paper 1 uses an unseen resource. Practise reading unfamiliar data quickly. IB rewards use of stimulus material — reference figures/tables explicitly.

Key Biogeochemical Cycles

CARBON CYCLE

CARBON CYCLE — KEY STORES & FLOWS

Store	Est. Size
Ocean (dissolved)	~38,000 GtC (largest)
Fossil fuels	~4,000 GtC
Terrestrial biosphere	~2,500 GtC (soil + vegetation)
Atmosphere (CO_2)	~870 GtC

A Photosynthesis CO_2 → organic C (plants)

B Respiration Organic C → CO_2 (all organisms)

C Decomposition Dead matter → CO_2 + nutrients

D Combustion Fossil C + biomass → CO_2

E Ocean uptake CO_2 dissolves in seawater

F Limestone formation Marine organisms → $CaCO_3$

G Weathering $CaCO_3$ + rain → dissolved Ca + CO_2

H Fossilisation Organic matter → coal/oil/gas (millions of yrs)

💡 **EXAM TIP:** Know BOTH natural and anthropogenic carbon fluxes. Burning fossil fuels releases carbon stored over millions of years in decades — this is why the cycle is unbalanced.

NITROGEN CYCLE

NITROGEN CYCLE — KEY PROCESSES

1 N_2 Fixation N_2 → NH_3 (bacteria: Rhizobium, Azotobacter; lightning)

2 Nitrification NH_3 → NO_2^- → NO_3^- (Nitrosomonas, Nitrobacter)

3 Assimilation Plants absorb NO_3^- → organic N (amino acids, proteins)

4 Consumption Animals eat plants → organic N transferred up food chain

5 Decomposition Dead organisms → NH_3/NH_4^+ (ammonification)

6 Denitrification NO_3^- → N_2 (anaerobic bacteria return N to atmosphere)

7 Leaching NO_3^- washed into waterways → eutrophication risk

8 Human Inputs Haber-Bosch process → synthetic fertilisers; large N inputs

Eutrophication Process (linked to N cycle)

Excess N/P enters water → Algal bloom → Light blocked → Plants die, decomposed → O_2 depletion (hypoxia) → Fish kills, dead zone

⚠️ **COMMON MISTAKE:** In eutrophication, it is DECOMPOSERS (bacteria) consuming oxygen, not algae. Fish die from oxygen depletion, not from the algae directly.

✅ Final Revision — Night Before Checklist

📝 CORE CONCEPT SUMMARIES

Must-Know Definitions

- Open/closed/isolated system + examples
- GPP, NPP, and calculation
- Simpson's Diversity Index formula
- Positive vs negative feedback loops
- In-situ vs ex-situ conservation
- Eutrophication process (all steps)
- Enhanced greenhouse effect mechanism
- Sustainable development definition (Brundtland)
- Ecocentric / Anthropocentric / Technocentric EVS

Must-Know Calculations

- NPP = GPP − R
- Energy efficiency = $(T_{n+1} / T_n) \times 100$
- Simpson's D = $1 - \Sigma[n(n-1)/N(N-1)]$
- % change calculations in data questions

Must-Know Case Studies

- Aral Sea — water/irrigation disaster
- Virunga/Mountain Gorilla — conservation
- Atlantic Cod collapse — overfishing
- Yellowstone wolves — trophic cascade
- Great Barrier Reef — climate/coral
- Montreal Protocol — ozone success
- Paris Agreement — climate governance
- Millennium Seed Bank — ex-situ conservation
- Germany Energiewende — energy transition

☐ NIGHT BEFORE EXAM CHECKLIST

- ☐ I can draw and label the carbon cycle from memory
- ☐ I can draw and label the nitrogen cycle from memory
- ☐ I can draw a food web with 5+ species and label trophic levels
- ☐ I can explain positive AND negative feedback with examples
- ☐ I know at least 1 case study per major topic
- ☐ I understand the difference between GPP and NPP
- ☐ I can calculate Simpson's D from a data table
- ☐ I know all IB command terms and what each requires
- ☐ I can evaluate in-situ vs ex-situ conservation (pros/cons)
- ☐ I can explain eutrophication step-by-step
- ☐ I know the difference between ozone depletion and climate change
- ☐ I can name 3+ mitigation strategies and evaluate each
- ☐ I have read the exam paper structure and time allocation
- ☐ I have reviewed my mock exam mistakes
- ☐ I know common mistakes to avoid (see p.17)

💡 **EXAM TIP:** The night before: review this checklist. Tick what you know. Spend the last hour on unticked items only. Sleep well — recall improves with sleep!

You've got this. 🌿

ESS rewards students who think critically, evaluate thoroughly, and support arguments with examples.

Be precise. Be balanced. Be curious.

Target: 7 ✓